

Successive ionisation energies (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to deduce the group of an element from its successive ionisation data.

In the previous [section S1.3.6 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculating-ionisation-energies-hl-id-44044/) you learned that electrons can be removed from an atom and that the energy required to do this is the ionisation energy. But is it possible to remove more than one electron if enough energy is supplied? Would doing so require the same amount of energy or more energy? Rather than just removing one electron, it is possible to continue to remove electrons as long as energy is supplied. These are known as successive ionisation energies and as you will learn in this section, we can use successive ionisation data to determine to which group an element belongs.

Successive ionisation energies

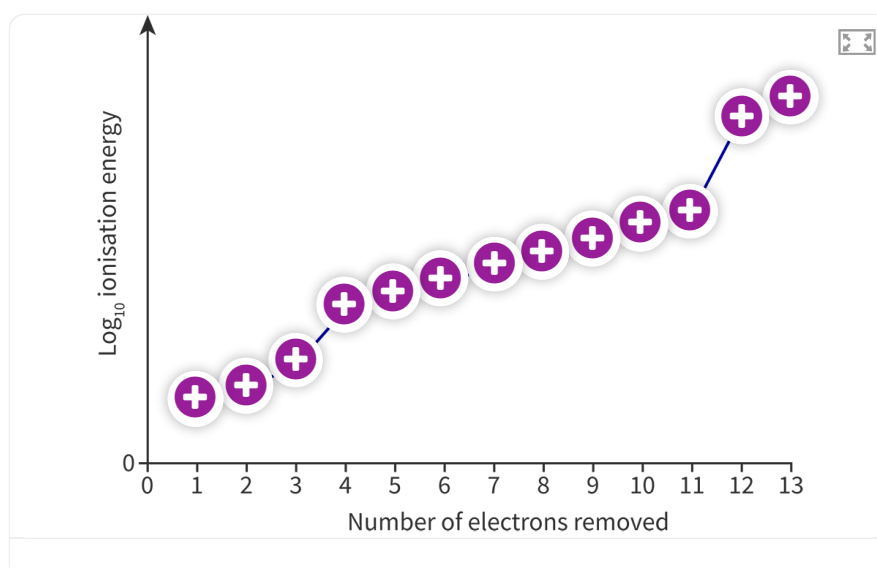
Successive ionisation refers to the process of removing successive electrons from an atom or positive ion. Taking aluminium as an example, we can write equations that represent the successive ionisation energies (**Table 1**). Note that after the first ionisation energy, each successive ionisation removes an electron from an increasingly positive ion. This has an important effect on the value of the ionisation energy and will be discussed later in the section.

Table 1. Equations for the successive ionisation of aluminium.

Ionisation	Equation
1st	$\text{Al(g)} \rightarrow \text{Al}^{\text{+}}(\text{g}) + \text{e}^{-}$
2nd	$\text{Al}^{\text{+}}(\text{g}) \rightarrow \text{Al}^{\text{2+}}(\text{g}) + \text{e}^{-}$
3rd	$\text{Al}^{\text{2+}}(\text{g}) \rightarrow \text{Al}^{\text{3+}}(\text{g}) + \text{e}^{-}$
4th	$\text{Al}^{\text{3+}}(\text{g}) \rightarrow \text{Al}^{\text{4+}}(\text{g}) + \text{e}^{-}$
5th	$\text{Al}^{\text{4+}}(\text{g}) \rightarrow \text{Al}^{\text{5+}}(\text{g}) + \text{e}^{-}$
6th	$\text{Al}^{\text{5+}}(\text{g}) \rightarrow \text{Al}^{\text{6+}}(\text{g}) + \text{e}^{-}$

Ionisation	Equation
7th	$\text{Al}^{6+}(\text{g}) \rightarrow \text{Al}^{7+}(\text{g}) + \text{e}^{-}$
8th	$\text{Al}^{7+}(\text{g}) \rightarrow \text{Al}^{8+}(\text{g}) + \text{e}^{-}$
9th	$\text{Al}^{8+}(\text{g}) \rightarrow \text{Al}^{9+}(\text{g}) + \text{e}^{-}$
10th	$\text{Al}^{9+}(\text{g}) \rightarrow \text{Al}^{10+}(\text{g}) + \text{e}^{-}$
11th	$\text{Al}^{10+}(\text{g}) \rightarrow \text{Al}^{11+}(\text{g}) + \text{e}^{-}$
12th	$\text{Al}^{11+}(\text{g}) \rightarrow \text{Al}^{12+}(\text{g}) + \text{e}^{-}$
13th	$\text{Al}^{12+}(\text{g}) \rightarrow \text{Al}^{13+}(\text{g}) + \text{e}^{-}$

Interactive 1 shows a graph of the successive ionisation energies of aluminium. Move your mouse over each point to see from which sublevel the electron is being removed. From which sublevel do we start removing electrons? Is it from the sublevel closest to the nucleus or furthest from the nucleus?



Interactive 1. The Successive Ionisation Energies of Aluminium.

 More information for interactive 1

An interactive graph exhibits the ionisation energies of aluminium. The x-axis represents the number of electrons removed from aluminium, ranging from 0 to 13, while the y-axis shows the log10 of ionisation energy.

The graph features an increasing trend, with plus signs marking interactive hotspots along the line. Selecting these hotspots reveals the subshell from which the electrons are removed.

The number of electrons removed and their corresponding sublevels are listed in a table below:

This graph helps users understand that electrons removed from inner orbitals have higher ionisation energies, while those from outer orbitals require less energy to be removed.

Number of electrons removed	Sublevel
1	3p
2	3s
3	3s
4	2p
5	2p
6	2p
7	2p
8	2p
9	2p
10	2s
11	2s
12	1s
13	1s

The order in which electrons are removed is from the higher energy sublevels first – so the first electron is removed from the 3p sublevel, and the last electron is removed from the 1s sublevel. When electrons are removed, they are always removed from the highest energy sublevel first.

From the graph we can also see that successive ionisation energies show an increase. The reason for this is that after the first ionisation, the electrons are being removed from an increasingly positive ion. This leads to an increase in the electrostatic attraction between the nucleus and the remaining electrons and an increase in the ionisation energy.

Making connections

Before you proceed with the rest of this section, it is helpful to learn about the groups of the periodic table; this is covered in [section S3.1.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/>).

Successive ionisation energies provide evidence as to the electron configuration of an atom, in particular with respect to the number of valence electrons. Recall that valence electrons are those electrons located in the outermost shell of an atom. Elements with the same number of valence electrons are placed in the same group on the periodic table. The periodic table is divided into vertical groups numbered from 1–18. The main group elements are those located in groups 1 and 2 and 13 to 18. For these elements, **Figure 1** tells us how many valence electrons an atom has.

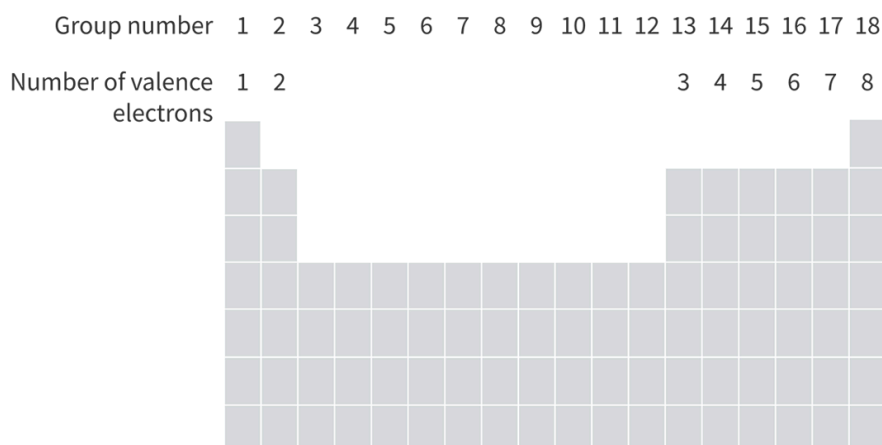


Figure 1. The number of valence electrons in each main group of the periodic table.

 More information for figure 1

The image shows a periodic table of elements, highlighting the main groups which indicate the number of valence electrons. The main groups are marked as vertical columns, and the applicable groups range from 1 to 2 and 13 to 18, which are shaded differently to show their significance. This differentiation helps in understanding the electron configuration by emphasizing the vertical arrangement of elements having the same number of valence electrons. Light grey squares represent the elements in the table, with darker shading for the main groups. The periodic table is structured with elements arranged in order of increasing atomic number.

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Figure 2 shows the successive ionisation energies of aluminium again, but this time showing the main energy level to which each electron belongs. What do you notice about the graph? Is the increase in ionisation energy constant or are there large increases every so often? And what do these large increases mean?

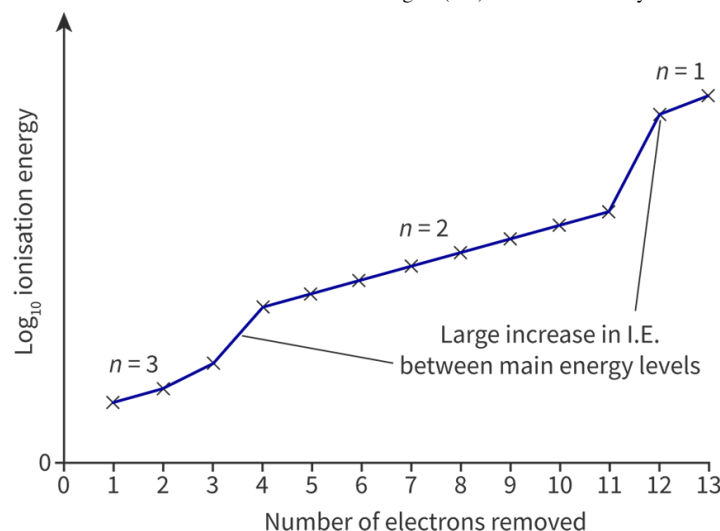


Figure 2. The successive ionisation energies of aluminium.

More information for figure 2

The graph illustrates the successive ionisation energies of aluminium. The X-axis represents the electron number removed, ranging from 1 to approximately 13, while the Y-axis represents the ionisation energy in arbitrary units. The graph shows a series of step-like increases with two significant jumps in energy levels. The first major increase occurs after the removal of the third electron, while the second substantial rise happens after the removal of the eleventh electron. These large increases indicate electrons being removed from different energy levels, with the 4th and 12th electrons notably requiring more energy as they belong to lower energy levels than previous ones. This pattern supports the concept of energy levels within the atom.

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Two large increases can be seen in the graph; one after the 3rd electron is removed and another after the 11th electron is removed. These large increases correspond to the 4th and 12th electrons being removed from different energy levels to the previous electrons. The 3rd electron is removed from $n = 3$ whereas the 4th electron is removed from $n = 2$. And the 11th electron is removed from $n = 2$ whereas the 12th electron is removed from $n = 1$. These large increases are further evidence for the existence of energy levels in the atom.

As mentioned earlier we can also use these large increases to determine to which group an element belongs, based on the number of valence electrons an atom has. The first large increase in a graph of successive ionisation energies occurs after the removal of the valence electrons. To determine the group that the element belongs to, we need to count the number of electrons before this large increase, which tells us the number of valence electrons in the atom and to which group the element belongs. So for aluminium this occurs after the 3rd electron is removed which tells us that aluminium has three valence electrons and is located in group 13 of the periodic table.

Data for successive ionisation energies can also be represented in table form. **Table 2** shows the successive ionisation data for the first four ionisations of aluminium.

Table 2. Successive ionisation data for the first four ionisations of aluminium.

Ionisation	1st	2nd	3rd	4th
Ionisation energy / kJ mol^{-1}	577	1820	2740	11508

Once again, we see the large increase in ionisation energy occurs after the third electron is removed, indicating that the fourth electron is removed from an energy level closer to the nucleus. The large increase after the third electron is removed tells us once again that aluminium is located in group 13 of the periodic table.

Worked example 1

Figure 3 shows all the successive ionisation energies of an element.

- How many occupied energy levels does the atom contain?
- From which energy level is the first electron removed?
- To which group of the periodic table does the element belong?
- What is the identity of the element?

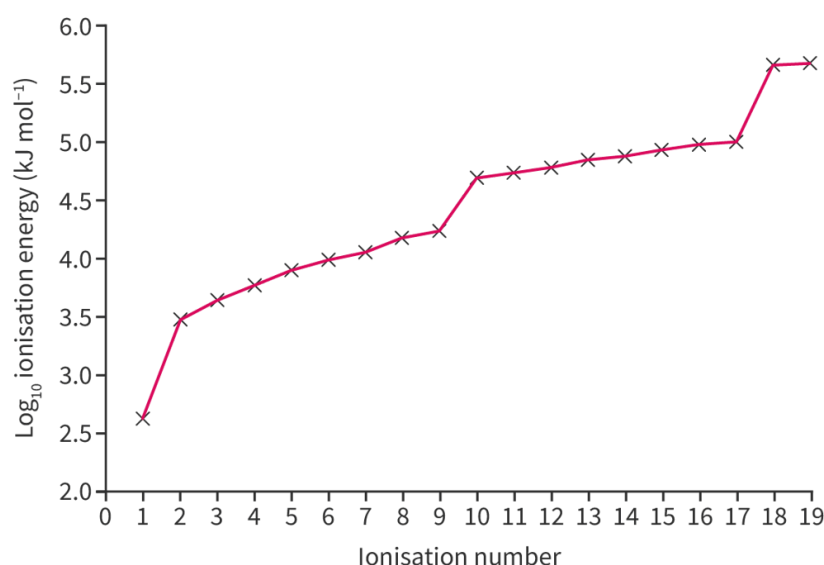


Figure 3. Successive ionisation energies of an element.

More information for figure 3

The graph displays the successive ionisation energies of an element on a logarithmic scale. The X-axis represents the ionisation number, ranging from 0 to 19. The Y-axis shows the log base 10 of ionisation energy in kJ mol^{-1} , ranging from 2.0 to 6.0. Data points are marked with an 'X' and connected by a line. The graph indicates an overall upward trend, with slight variations in slope at various points. Notably, there are more pronounced increases in ionisation energy around the ionisation numbers 9 to 10, and 17 to 18, suggesting higher energy requirements to remove these electrons. Between these points, the increases are more gradual, indicating less energy is needed to remove those electrons.

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- The number of occupied energy levels can be determined by the number of 'jumps' or large increases in ionisation energy. There are three large increases in ionisation energy so there are four occupied energy levels.

- (b) The first electron is removed from the $n = 4$ energy level.
- (c) There is one electron removed before the first big increase in ionisation energy, so the element has one valence electron and is located in group one of the periodic table.
- (d) There are a total of 19 electrons removed so the atom must have 19 protons. The element with 19 protons is potassium ($Z = 19$).

Worked example 2

Data for the successive ionisation energies of an unknown element is shown in **Table 3**. Deduce how many valence electrons the unknown element has and to which group of the periodic table it belongs.

Table 3. Successive ionisation data for an unknown element.

Ionisation	1st	2nd	3rd	
Ionisation energy / kJ mol^{-1}	590	1145	4912	6

To determine the number of valence electrons and group number, we look for the large increase in ionisation energy. The large increase occurs after the second electron is removed. The ionisation energy increases from 1145 kJ mol^{-1} to 4912 kJ mol^{-1} because the third electron is removed from an energy level closer to the nucleus. Therefore, the element has two valence electrons and is located in group 2 of the periodic table.

Making connections

Subtopic S3.1 (HL) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>) covers how patterns of successive ionisation energies of transition elements help to explain the variable oxidation states of these elements?

In this activity, you will plot a graph of successive ionisation energies and explain to which group the element belongs.

Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

In this activity, you will choose an element and plot a graph of its successive ionisation energies.

1. Click on the link below.

<https://www.webelements.com/atoms.html> 
(<https://www.webelements.com/atoms.html>)

2. Choose an element from group 15—18 in period three of the periodic table and click on that element.
3. Write the full electron configuration of your chosen element.
4. Find the successive ionisation data for your chosen element and record it in table form.
5. Use a spreadsheet such as Google Sheets to plot a graph of the number of electrons removed (on the x-axis) against ionisation energy (on the y-axis).
6. Label the electrons with the main energy sublevel they belong to. Label the large increases in ionisation energy and explain each increase.
7. Explain how the graph confirms to which group your chosen element belongs.